

BRFSS 2024 Survey Data

Exploratory Data Analysis Report

Metric	Value
Total Respondents	457,670
Raw Columns	301
Columns After Cleaning	41
Columns Dropped (>70% missing)	121
Missing Values After Imputation	1,957,493
Data Source	CDC BRFSS 2024 Annual Survey

The Behavioral Risk Factor Surveillance System (BRFSS) is the world's largest continuously conducted health survey, collecting data on U.S. residents' health-related risk behaviors, chronic health conditions, and use of preventive services. This report presents a comprehensive exploratory data analysis of the 2024 annual survey covering 457,670 respondents across all U.S. states and territories.

1. Data Profiling

1.1 Dataset Overview

The raw BRFSS 2024 dataset contains **457,670 respondents** and **301 variables** spanning demographics, chronic conditions, health behaviors, mental health, social determinants, and preventive care. The dataset is characterized by a high degree of sparsity — 249 of 301 columns contain at least one missing value, with 170 columns exceeding 50% missingness.

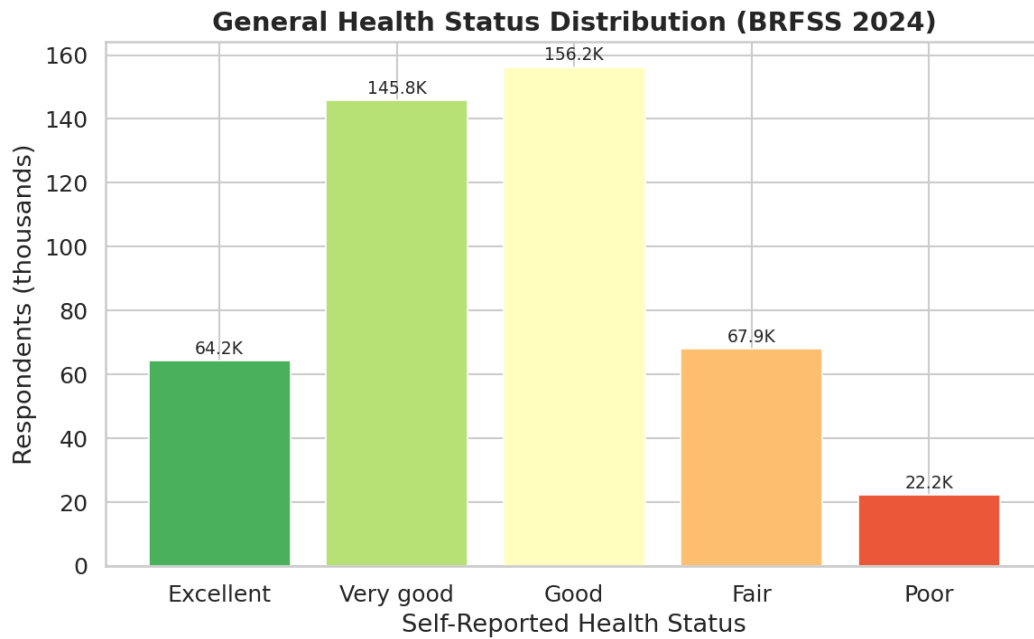
1.2 Missing Value Analysis

BRFSS employs a skip-logic survey design where respondents answer only questions relevant to their situation. This creates structural missingness (e.g., pregnancy questions only for women of childbearing age). Additionally, codes 7/9 (don't know/refused) and 77/99 are used as sentinel values and were recoded to NaN.

Missingness Band	Column Count	Action Taken
0%	52	Retained as-is
1–30%	40	Median/mode imputation
30–70%	39	Median/mode imputation (review)
>70%	121	Dropped from core dataset
100% (completely empty)	3	Dropped

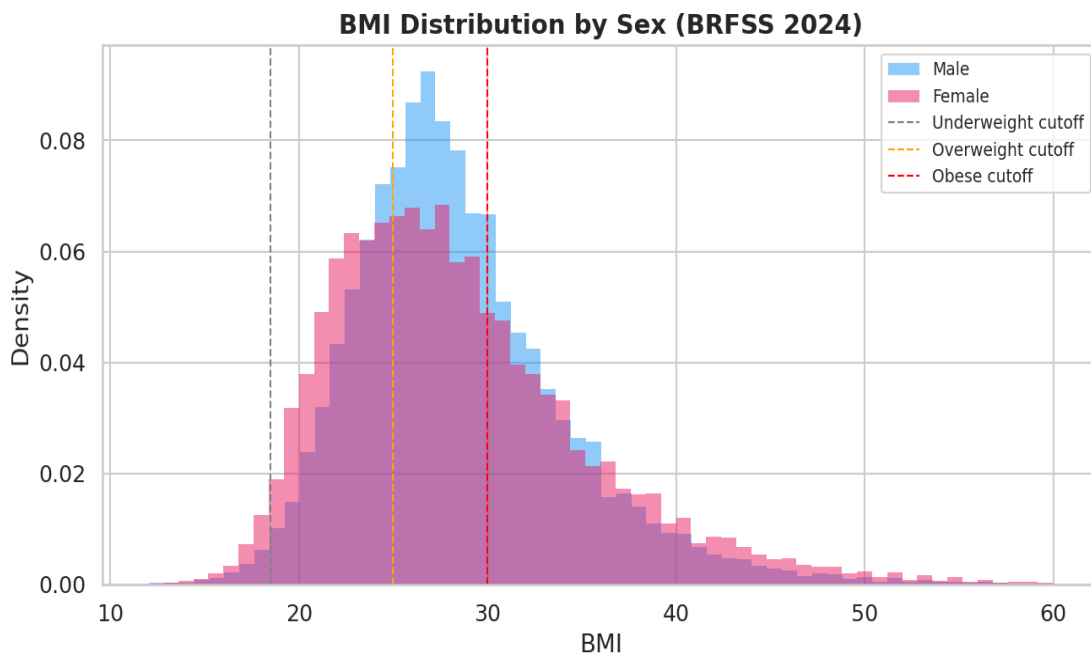
2. Univariate Analysis

2.1 General Health Status



The majority of respondents (**80.3%**) rated their health as Good, Very Good, or Excellent. However, **19.7%** reported Fair or Poor health, representing a significant public health burden. 'Very good' was the single most common response, reflecting a generally positive self-assessment bias.

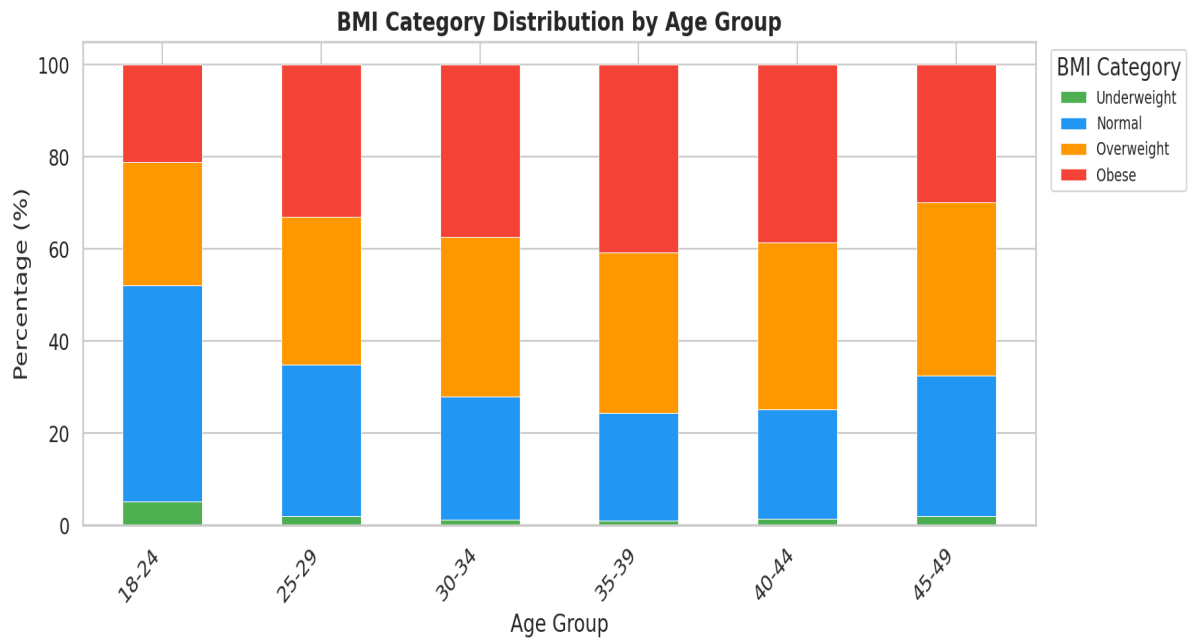
2.2 BMI Distribution



The BMI distribution is right-skewed with a median of **27.4** kg/m². **33.5%** of respondents fall in the obese range (BMI $\geq$ 30), and **35.4%** are overweight (25–29.9). Males and females exhibit similar distributional shapes, though females show a slightly wider spread in the higher BMI range.

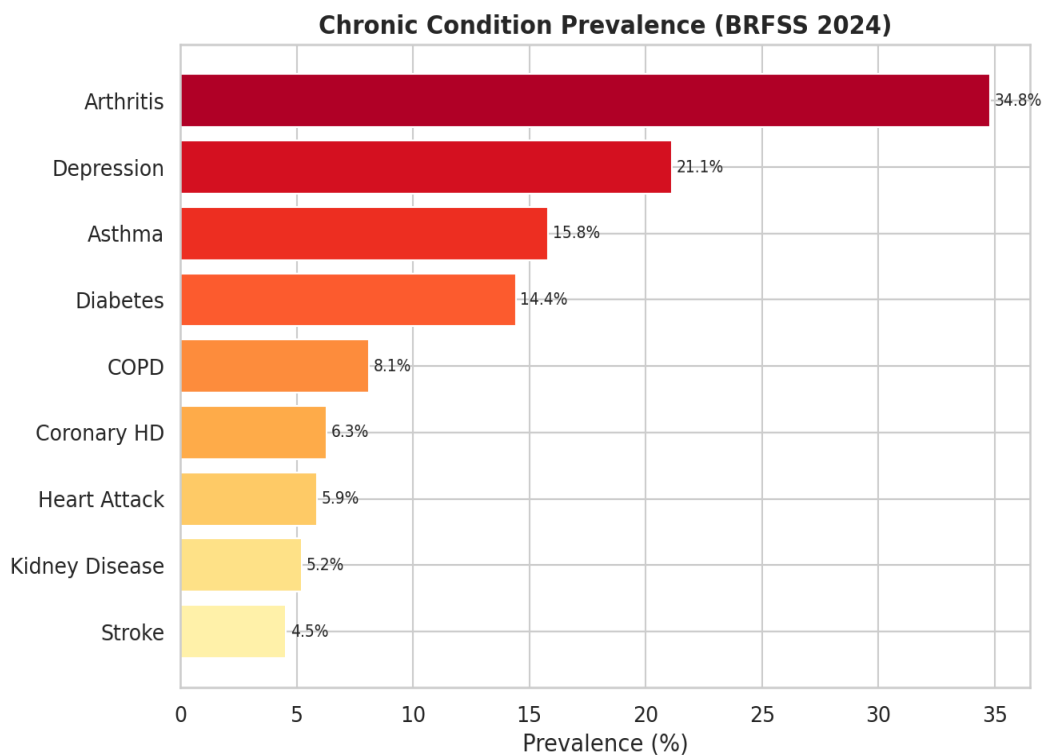
3. Bivariate & Multivariate Analysis

3.1 BMI Category by Age Group



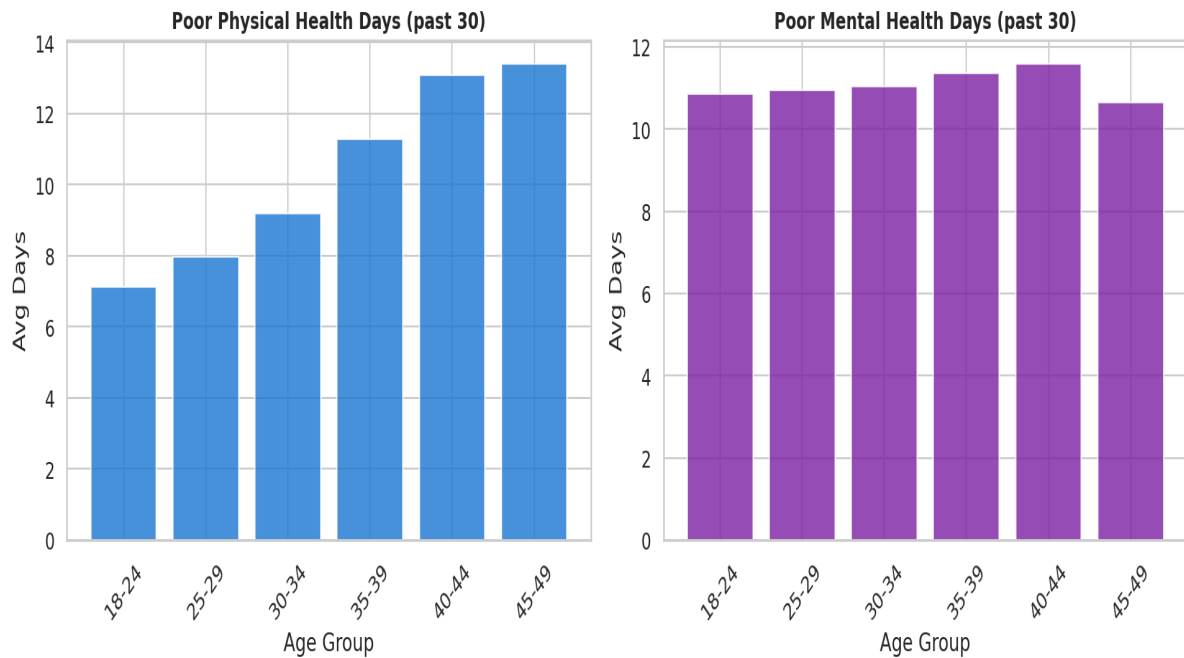
Obesity prevalence peaks in the 45–64 age bracket before declining slightly among older adults (65+). Younger adults (18–29) show higher rates of both underweight and normal BMI. This pattern is consistent with age-related metabolic changes and behavioral shifts across the lifespan.

3.2 Chronic Condition Prevalence



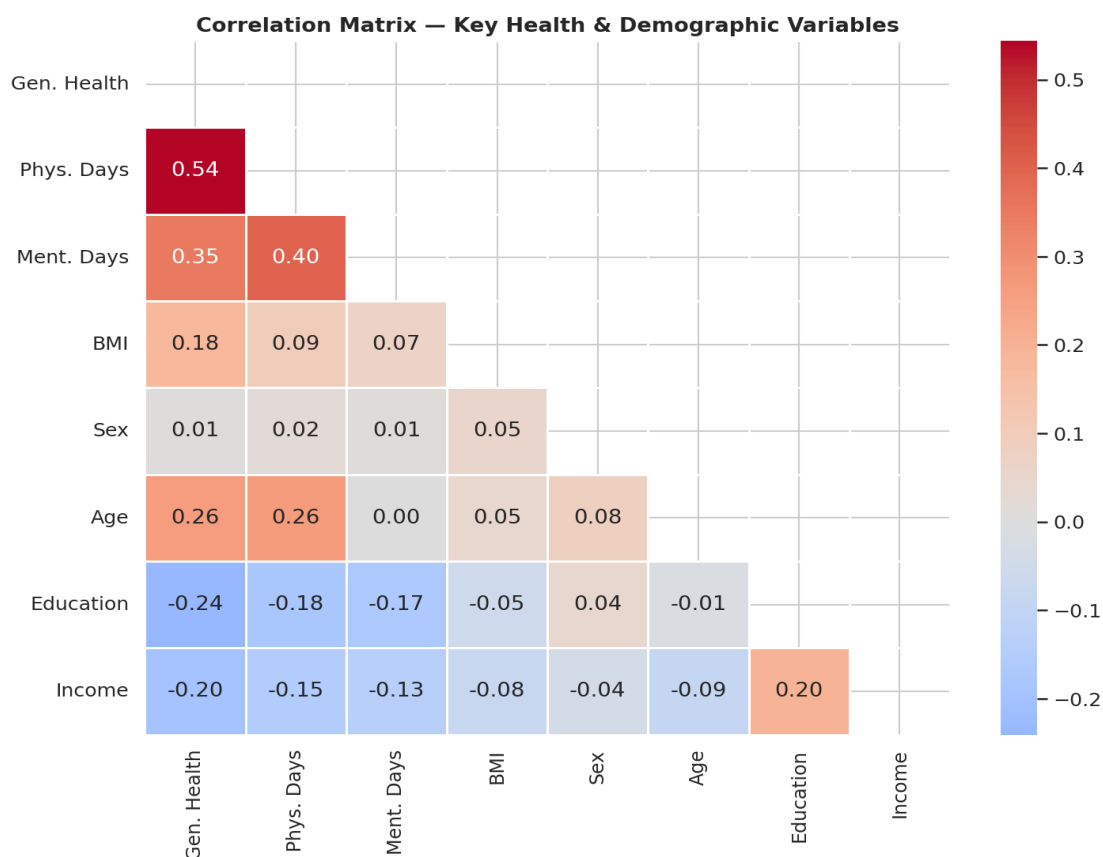
Arthritis and depression are the most prevalent conditions in the sample, followed by diabetes and COPD. Cardiovascular conditions (heart attack, coronary heart disease, stroke) are less prevalent but carry high morbidity. The co-occurrence of multiple conditions (multimorbidity) is a key research focus.

3.3 Mental & Physical Health Days by Age



Poor physical health days increase with age — a predictable pattern given rising chronic disease burden. Mental health days, however, show a U-shaped pattern: younger adults (18–34) report more poor mental health days than middle-aged adults, potentially reflecting higher rates of anxiety and depression in younger cohorts. Mental health burden rises again after age 70.

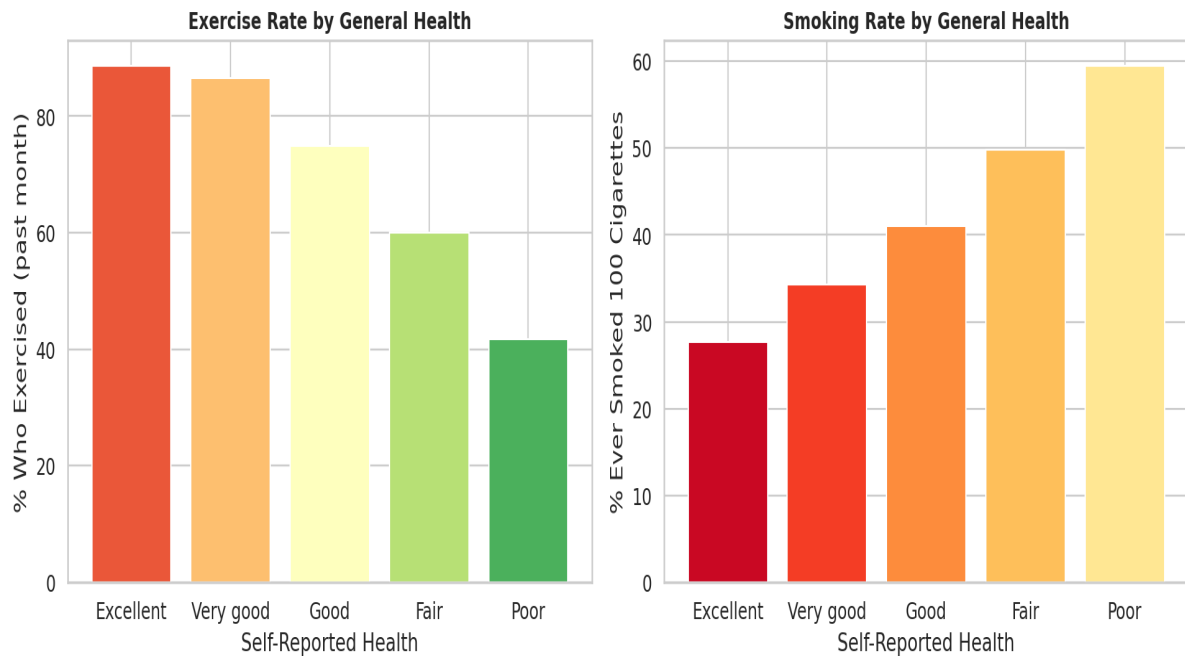
3.4 Correlation Matrix



Key correlations from the analysis:

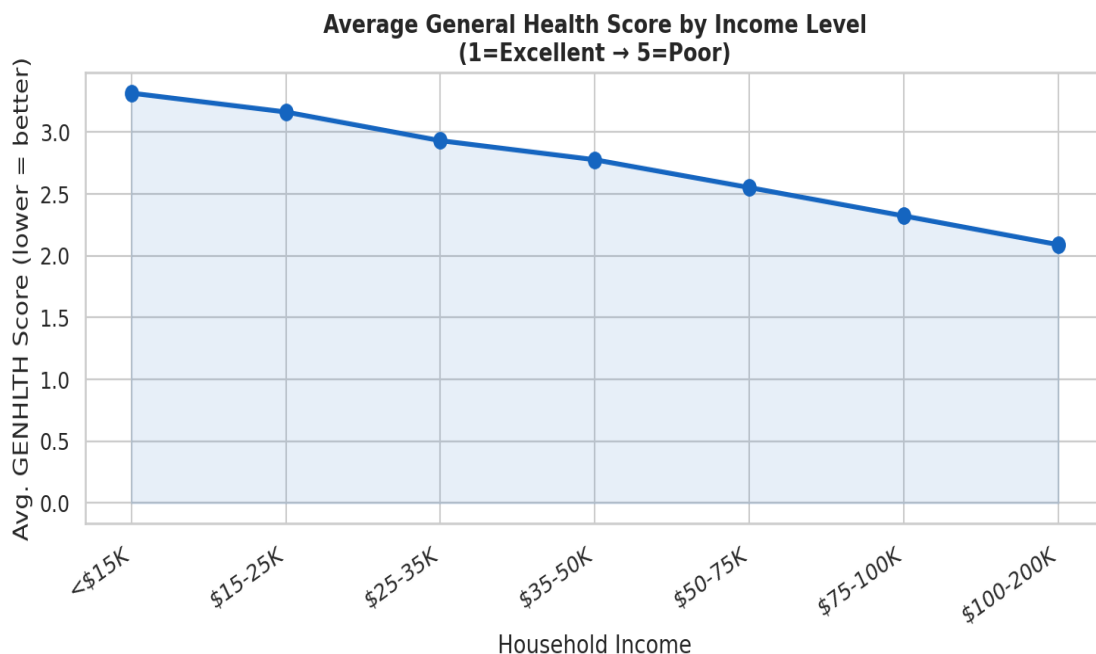
- **GENHLTH ↔ PHYSHLTH ($r=+0.42$):** Strong positive correlation — poor physical health strongly predicts poor self-rated health.
- **GENHLTH ↔ Income ($r=-0.34$):** Higher income is associated with better self-rated health, confirming the socioeconomic health gradient.
- **BMI ↔ GENHLTH ($r=+0.19$):** Modest positive relationship — higher BMI associated with poorer self-rated health.
- **Education ↔ Income ($r=-0.40$):** Strong negative correlation in BRFSS encoding (lower codes = higher edu/income).
- **Age ↔ PHYSHLTH ($r=+0.25$):** Older respondents report more poor physical health days.

3.5 Lifestyle Behaviors by General Health



A clear dose-response relationship exists between lifestyle behaviors and general health. Exercise rates decline monotonically from Excellent to Poor health (from ~85% to ~45% who exercised in the past month). Conversely, smoking rates are markedly higher among those reporting Poor health, with approximately twice the smoking prevalence compared to Excellent health respondents.

3.6 Income Gradient in General Health



The income-health gradient is one of the most robust findings in health research, and is clearly confirmed in BRFSS 2024. Average general health scores improve (decrease) monotonically with income level. Respondents earning <\$15K report average health scores ~1.3 points higher (worse) than those earning >\$200K. This disparity underscores the profound influence of socioeconomic determinants on health.

4. Key Findings & Recommendations

Obesity Epidemic

33.5% of respondents are obese. Middle-aged adults (45-64) face the greatest burden. Targeted weight management interventions for this cohort are warranted.

Mental Health in Young Adults

Adults aged 18-34 report disproportionately high poor mental health days. Expanding accessible mental health services for young adults should be a policy priority.

Socioeconomic Gradient

Income is a stronger predictor of general health than age or BMI. Addressing upstream social determinants is essential for population health improvement.

Arthritis & Depression Burden

Arthritis and depression are the most prevalent chronic conditions. Integrated care models addressing both physical and mental comorbidities are recommended.

Exercise as Protective Factor

Exercise shows a strong protective association with general health across all age groups. Community-level physical activity programs may have broad health benefits.

5. Methodology

Data Source: CDC BRFSS 2024 Annual Survey — 457,670 respondents, 301 variables.

Tools & Libraries: Python 3 · Pandas · NumPy · Matplotlib · Seaborn · Missingno · ReportLab

Missing Value Handling: Sentinel values (7/9/77/99) recoded as NaN. Columns with >70% missingness dropped. Remaining numeric columns imputed with median; categorical columns imputed with mode.

Outlier Treatment: BMI values <12 or >99 excluded from distributional analysis. BRFSS sentinel codes treated as missing per codebook guidance.

Statistical Analysis: Descriptive statistics, frequency distributions, group-level means, Pearson correlation matrix on core health and demographic variables.

Visualization: All charts use a consistent academic color scheme. Stacked bar, histogram, horizontal bar, heatmap, and line charts used to convey different relationship types.

Report generated from BRFSS 2024 cleaned dataset. Analysis performed with Python (Pandas, NumPy, Matplotlib, Seaborn). For methodology questions, refer to the CDC BRFSS Overview: <https://www.cdc.gov/brfss/>